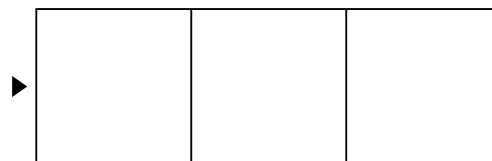
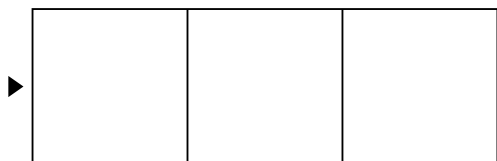
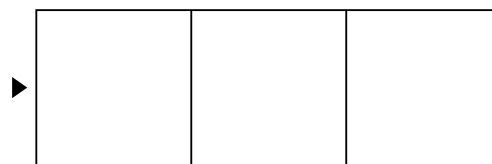
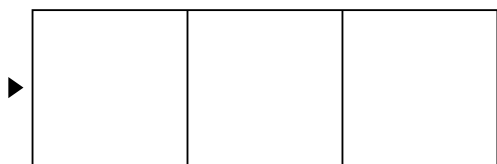
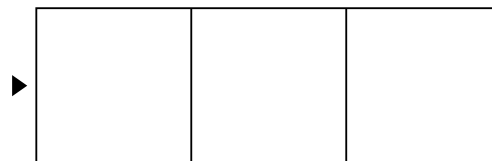
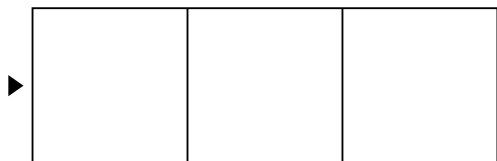
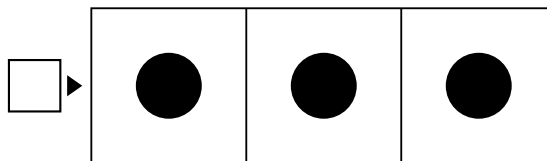
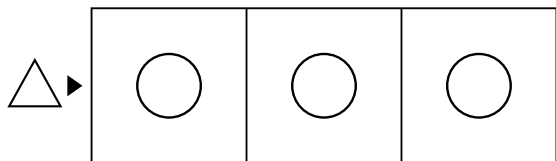
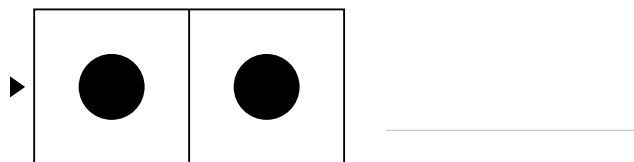
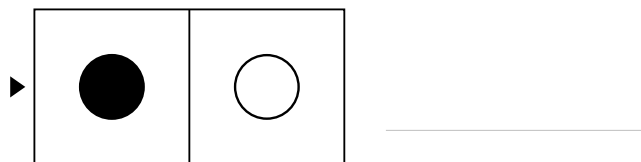
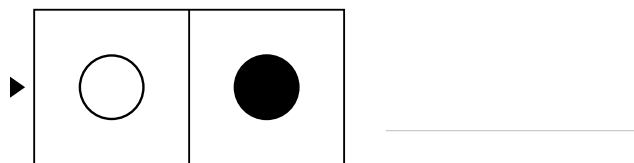
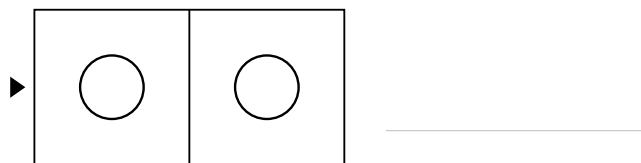
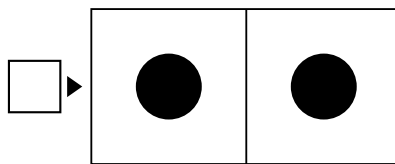
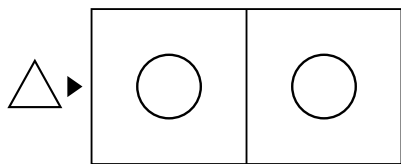
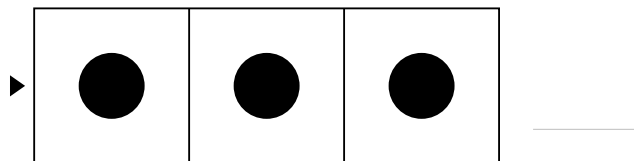
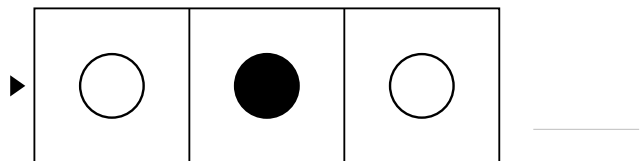
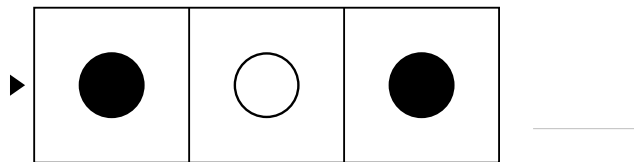
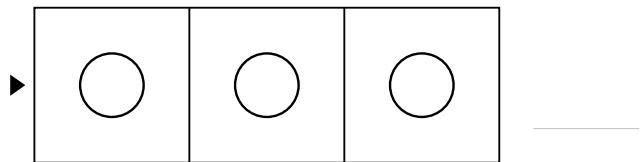
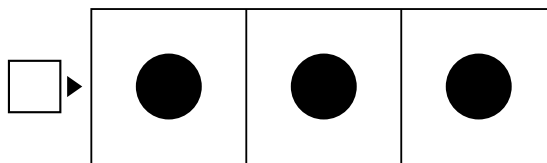
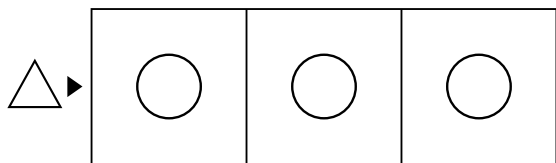


Keep the arrow at the left. The sender hides a picture and its row. The changer turns over zero or one counter in secret. Only the final row crosses the folder; the receiver uses the key to find the picture.

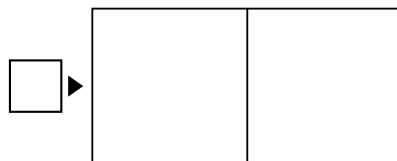
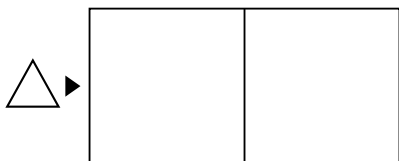
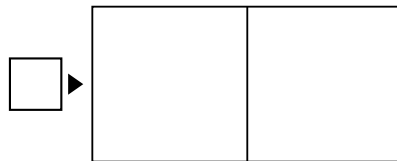
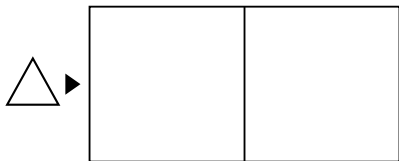
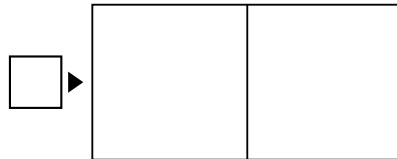
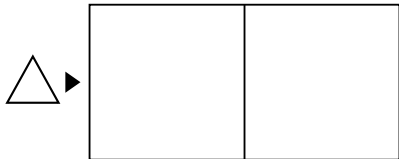
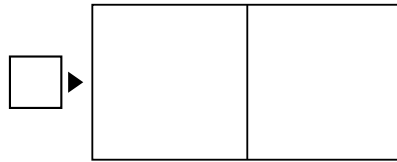
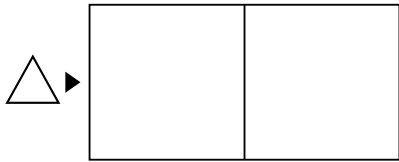
Problem 1: Send both pictures with this key many times. Can the changer fool the receiver?



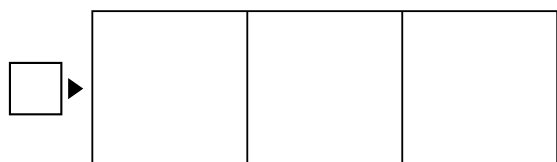
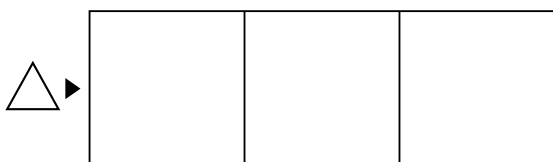
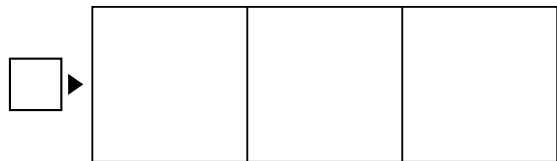
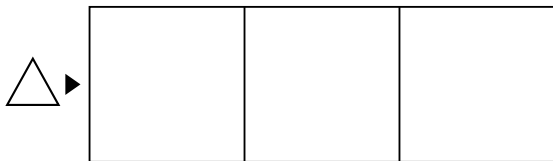
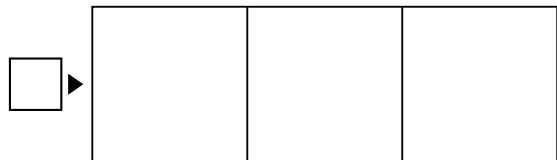
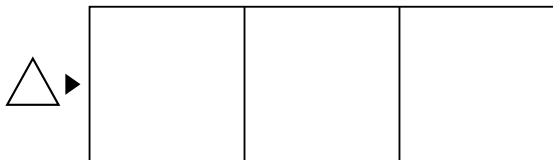
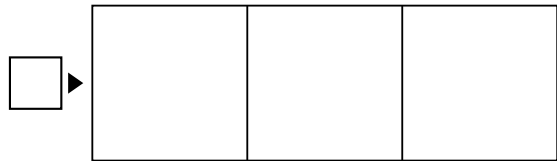
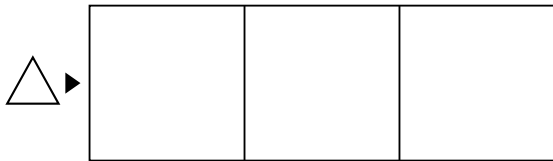
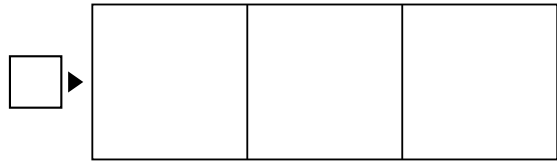
Problem 2: Draw every possible picture beside each final row. Which key lets the changer fool the receiver?



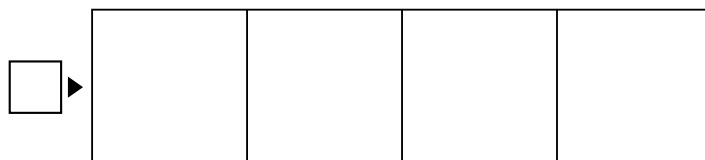
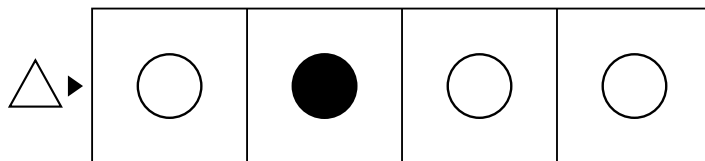
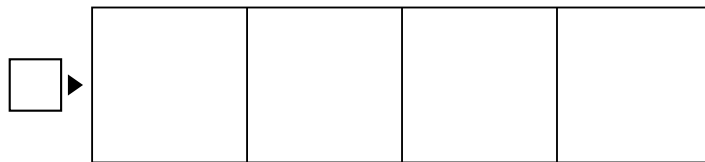
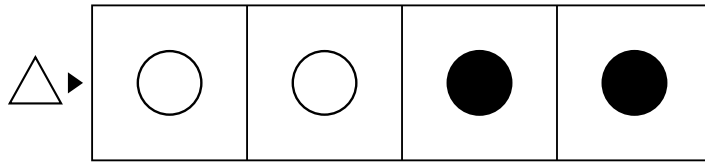
Problem 3: Make two two-counter rows for the pictures. Can you keep the changer from fooling the receiver?



Problem 4: Find every three-counter key that always tells the two pictures apart; use more paper if needed. Swapping the pictures makes a different key.



Problem 5: For each key, find every square row that always works. Use both kinds of counter in each row, and use more paper if needed.



Problem 6: Can three pictures share a three-counter key that always works?
Make one, or show why it cannot be done.

